

QE3

- Forfall Ingen forfallsdato
- Poeng 8
- Spørsmål 5
- Tilgjengelig 29. okt. i 15.15 - 29. okt. i 18:00 2 timer og 45 minutter
- Tidsbegrensning Ingen
- Tillatte forsøk Ubegrenset

Denne quizen ble låst 29. okt. i 18:00.

Forsøkshistorikk

	Forsøk	Tid	Resultat
GJELDENDE	Forsøk 2	13 minutter	2 av 8
SISTE	Forsøk 2	13 minutter	2 av 8
	Forsøk 1	24 minutter	0 av 8

Resultat for dette forsøket: 2 av 8

Innlevert 29. okt. i 17.27

Dette forsøket tok 13 minutter

Riktig svar



Spørsmål 1

2 / 2 poeng

DC motor speed control 3 . Consider a shunt DC motor (SHDC) . It is subjected to a constant load (τ_L) . The supply voltage should be controlled by means of PWM with $T_A/T_s = 200$. Armature resistance (R_A) = 0.88 Ohm . Armature inductance (L_A) = 30 mH . Field resistance (R_F) = 61.3 Ohm . Knee point field current ($i_{F,kp}$) = 2.91 A . Knee point induced armature voltage ($e_{A,kp}(0)$) = 112.9 Volt . Induced armature voltage reference motor speed ($\omega_M(0)$) = 56.6 rad/s . Induced armature voltage shape factor (m) = 0.10 . Motor angular speed (ω_M) = 34.4 rad/s . Load torque (τ_L) = 281.2 N*m . PWM on-voltage (u_{on}) = 322 Volt . What is the PWM on-time (T_{on}) in ms?

0,1035 (med marginen: 0,001)

Feil svar



Spørsmål 2

0 / 2 poeng

DC motor speed control 2 . Consider a permanent magnet DC motor (PMD) . It is subjected to a constant load (τ_L) . Armature supply voltage (u_{sA}) = 895 Volt . Armature resistance (R_A) = 5.51

Ohm . Motor angular speed (ω_M) = 38.8 rad/s . Load torque (τ_L) = 729.7 N*m . Use the smallest of the two possible K_{Mfi} values . What is the motor constant ($K_{M\phi}$) in Wb?

6,1116 (med marginen: 0,0611)

Ubesvart



Spørsmål 3

0 / 2 poeng

DC motor speed control 1 . Consider a separately excited DC motor (SEDC) . It is delivering a constant output power (P) . Armature resistance (R_A) = 1.77 Ohm . Field supply voltage (u_{sF}) = 521 Volt . Field resistance (R_F) = 113.5 Ohm . Knee point field current ($i_{F,kp}$) = 3.58 A . Knee point induced armature voltage ($e_{A,kp(0)}$) = 286.3 Volt . Induced armature voltage reference motor speed ($\omega_{M(0)}$) = 87.1 rad/s . Induced armature voltage shape factor (m) = 0.16 . Motor angular speed (ω_M) = 83.5 rad/s . Motor output power (P) = 7382 W . What is the armature supply voltage (u_{sA}) in Volt?

363,4058 (med marginen: 3,6341)

Ubesvart



Spørsmål 4

0 / 1 poeng

SEDC and SHDC motors 1 . Consider a shunt DC motor (SHDC) . Armature supply voltage (u_{sA}) = 859 Volt . Armature resistance (R_A) = 3.87 Ohm . Field resistance (R_F) = 264.7 Ohm . Knee point field current ($i_{F,kp}$) = 2.60 A . Knee point induced armature voltage ($e_{A,kp(0)}$) = 538.3 Volt . Induced armature voltage reference motor speed ($\omega_{M(0)}$) = 80.8 rad/s . Induced armature voltage shape factor (m) = 0.21 . Motor angular speed (ω_M) = 42.1 rad/s . What is the motor armature current (i_A) in A?

136,5891 (med marginen: 1,3659)

Ubesvart



Spørsmål 5

0 / 1 poeng

SEDC and SHDC motors 1 . Consider a shunt DC motor (SHDC) . Armature supply voltage (u_{sA}) = 389 Volt . Armature resistance (R_A) = 5.89 Ohm . Field resistance (R_F) = 123.1 Ohm . Knee point field current ($i_{F,kp}$) = 3.33 A . Knee point induced armature voltage ($e_{A,kp(0)}$) = 268.9 Volt . Induced armature voltage reference motor speed ($\omega_{M(0)}$) = 89.2 rad/s . Induced armature voltage shape factor (m) = 0.16 . What is the stall torque (τ_{ST}) in N*m?



188,9329 (med marginen: 1,8893)

Quizresultat: 2 av 8